

- 2 A computer program is designed to simulate horses doing show jumping. In show jumping, horses jump over obstacles called fences. A horse successfully jumps a fence if it does not knock the fence down.

The program is written using Object-Oriented Programming (OOP).

The class `Horse` stores data about the horses.

Horse	
<code>Name : STRING</code>	stores the name given to the horse
<code>MaxFenceHeight : INTEGER</code>	stores the maximum height in cm that the horse can jump, for example 132
<code>PercentageSuccess : INTEGER</code>	stores the percentage chance of a horse not knocking down a fence, for example 70 represents a 70% chance of jumping a fence successfully
<code>Constructor()</code>	initialises <code>Name</code> , <code>MaxFenceHeight</code> and <code>PercentageSuccess</code> to its parameter values
<code>GetName()</code>	returns the name of the horse
<code>GetMaxFenceHeight()</code>	returns the maximum height the horse can jump
<code>Success()</code>	calculates and returns the percentage chance of a horse successfully jumping a specific fence

- (a) (i) Write program code to declare the class `Horse` and its constructor.

Do **not** declare the other methods.

Use your programming language's appropriate constructor.

All attributes must be private. If you are writing in Python, include attribute declarations using comments.

Save your program as **Question2_N24**.

Copy and paste the program code into part **2(a)(i)** in the evidence document.

[4]

- (ii) The get methods `GetName()` and `GetMaxFenceHeight()` each return the relevant attribute.

Write program code for the get methods.

Save your program.

Copy and paste the program code into part **2(a)(ii)** in the evidence document.

[3]